ZONAL E/E ARCHITECTURE COMPARATIVE STUDY REPORT

Legacy Point-to-Point vs 4-Zone Ethernet Architecture

Fleet Scope:

Freightliner Cascadia 126 (2020) | Western Star 49X (2021) | Mercedes-Benz Sprinter 519 CDI (2021)

Total ECUs modelled: 51 | Total connections modelled: 60

Tool: Zonal E/E Architecture Analyzer | AI Review: Groq Llama 3.3 70B

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Primary data source: DTNA SS-1033423, NHTSA public database

How to Read This Report

This report compares two ways of wiring electronics inside a commercial truck. The **old way (Legacy / Point-to-Point)** connects every Electronic Control Unit (ECU) directly to every other ECU it talks to using individual wires. The **new way (Zonal Architecture)** divides the truck into 4 physical regions. Each ECU connects to a nearby hub (Zone Controller) with a short wire. The 4 hubs then talk to each other over one Ethernet cable running along the truck. This tool models 3 vehicles, calculates the exact difference in wire length, weight, and complexity, and checks whether the human-drawn zone boundaries are mathematically optimal. A full abbreviation guide is on the next page.

Abbreviations and Full Forms

Every shortform used in this report is listed below. Refer to this page if you encounter an unfamiliar term.

Abbreviation	Full Form	Abbreviation	Full Form
ABS	Anti-lock Braking System	**J1939**	SAE Standard for heavy-duty vehicle CAN network
ACM	Aftertreatment Control Module	**LDW**	Lane Departure Warning
BCM	Body Control Module	**MCM**	Motor Control Module
CAN	Controller Area Network	**MPC**	Multi-Purpose Camera
CBZ	Cab Zone	**MSF**	Modular Switch Field
CGW	Central Gateway	**NHTSA**	National Highway Traffic Safety Administration
CHZ	Chassis Zone	**ONGUARD**	OnGuard Headway / Collision Controller
CPC	Common Powertrain Controller	**OTA**	Over-the-Air software update
CTP	Common Telematics Platform	**PTZ**	Powertrain Zone
DCMD	Door Controller — Driver Side	**SAE**	Society of Automotive Engineers
DCMP	Door Controller — Passenger Side	**SAM_CAB**	Signal Acquisition Module — Cab
DEF	Diesel Exhaust Fluid	**SAM_CH**	Signal Acquisition Module — Chassis
DPF	Diesel Particulate Filter	**SAS**	Steering Angle Sensor
DTCI	Daimler Truck Innovation Center India	**SCR**	Selective Catalytic Reduction
DTNA	Daimler Trucks North America	**SRS**	Supplemental Restraint System (Airbag)
E/E	Electrical / Electronic	**TCM**	Transmission Control Module
ECAS	Electronic Controlled Air Suspension	**TCO**	Tachograph
ECM	Engine Control Module	**TLM**	Telematics Module
ECU	Electronic Control Unit	**TPMS**	Tire Pressure Monitoring System
EPS	Electric Power Steering	**TSN**	Time-Sensitive Networking
ESC	Electronic Stability Control	**ZC**	Zone Controller
FCM	Forward Collision Module	**ZC-CAB**	Zone Controller — Cab Zone
FCU	Front Climate Control Unit	**ZC-CH**	Zone Controller — Chassis Zone
FRZ	Front Zone	**ZC-FR**	Zone Controller — Front Zone
HVAC	Heating Ventilation and Air Conditioning	**ZC-PT**	Zone Controller — Powertrain Zone
ICU	Instrument Cluster Unit	**100BASE-T1**	Automotive Ethernet — single-pair 100 Mbps

1. Executive Summary

The zonal E/E architecture analysis of the Freightliner Cascadia 126 (2020) reveals a 53.7% reduction in wire length, from 52.5m to 24.3m, resulting in a weight saving of 3.4 kg. The most important finding is the significant reduction in cross-zone runs from 16 to 3, indicating a more efficient zonal layout. The zone efficiency score of 7.0% suggests room for improvement in optimizing the zone boundaries.

1.1 Key Metrics — All Three Vehicles

Metric	Freightliner Cascadia 126	Western Star 49X	MB Sprinter 519 CDI
ECU Count	22	17	12
Legacy Wire Length	52.5m	56.5m	20.5m
Zonal Wire Length	24.3m	19.8m	13.9m
Wire Length Saved	**53.7%**	**65.0%**	**32.2%**
Cross-Zone Runs: Legacy	16	16	8
Cross-Zone Runs: Zonal	3	3	3
Harness Weight Saved	3.4kg	4.4kg	0.8kg
Zone Efficiency Score	7.0%	-4.9%	31.2%

Table 1: Key metrics — all three vehicles. Green = improvement. Red = legacy (higher is worse). Zone Efficiency = how well physical zones match communication-optimal groupings.

2. Background — What Is Zonal Architecture?

A modern truck has dozens of ECUs — small computers controlling everything from the engine and brakes to the dashboard display and door locks. Each ECU needs to send and receive data from other ECUs to do its job.

The Old Way — Legacy Point-to-Point Architecture

If ECU A needs to talk to ECU B, a dedicated wire is run between them. With 22 ECUs and 25 communication pairs on a Cascadia 126, this creates a dense web of wires crossing the full length of the truck. The engine controller (MCM) talking to the dashboard (ICU) needs a wire running ~5.5 meters. Multiply this across all 25 connections and you get 52.5 meters of wire and 6.3 kg of copper just for signal wiring. This is expensive to manufacture, difficult to route, and hard to repair.

The New Way — 4-Zone Ethernet Architecture

The truck is divided into 4 physical regions. Each ECU connects only to a nearby Zone Controller (ZC) with a short wire (0.5–1.5m). The 4 Zone Controllers then communicate over one Ethernet cable (100BASE-T1) running along the truck spine — about 7m total. When MCM needs to send data to ICU, it sends a message to ZC-PT, which forwards it over the backbone to ZC-CAB, which delivers it to ICU. No long individual wire needed. Result: 24.3m total wire, 2.9 kg, and only 3 cross-vehicle runs instead of 16.

Feature	Legacy Architecture	Zonal Architecture
Wiring principle	Direct wire between every communicating pair	Each ECU to local Zone Controller to Ethernet backbone
Cross-truck runs	One per communication pair (up to 16 long runs)	Only 3 backbone segments total
Wire length (Cascadia)	52.5 m	**24.3 m (-53.7%)**
Fault isolation	One broken wire can affect the whole system	Fault stays within the affected zone
Scalability	Complexity grows fast with each new ECU	Each new ECU adds only one short local wire
Industry status	Current standard on most trucks in service	Target for next-generation truck platforms

Table 2: Plain-language comparison of legacy and zonal architecture.

3. Vehicle Configurations and Zone Layouts

Three vehicles spanning the commercial vehicle spectrum were modelled. Each vehicle's ECUs were assigned to one of four physical zones based on their actual mounting location, cross-referenced with official DTNA and Mercedes-Benz service documentation.

3.1 Freightliner Cascadia 126

Total ECUs: 22 | **Connections:** 25 | **Source:** DTNA SS-1033423, NHTSA public database

Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	MCM, ACM, CPC, TCM	4
Chassis Zone	CHZ	ZC-CH	ABS, ECAS, TPMS, SAM_CHASSIS	4
Cab Zone	CBZ	ZC-CAB	ICU, SAM_CAB, CGW, FCU, SRS, MSF, PARKSMART, DCMD, DCOMP, TCO	10
Front Zone	FRZ	ZC-FR	MPC, ONGUARD, SAS, CTP	4

Table 3.1: Zone layout — Freightliner Cascadia 126

3.2 Western Star 49X

Total ECUs: 17 | **Connections:** 22 | **Source:** DTNA SS-1033423, Western Star Bodybuilder Manual Rev 3.1

Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	MCM, ACM, CPC, TCM, PTO	5
Chassis Zone	CHZ	ZC-CH	ABS, AXLE, TPMS, SAM_CH	4
Cab Zone	CBZ	ZC-CAB	ICU, CGW, SAM_CAB, FCU, MSF	5
Front Zone	FRZ	ZC-FR	ONGUARD, SAS, CTP	3

Table 3.2: Zone layout — Western Star 49X

3.3 Mercedes-Benz Sprinter 519 CDI

Total ECUs: 12 | **Connections:** 13 | **Source:** Mercedes-Benz XENTRY public diagnostic documentation

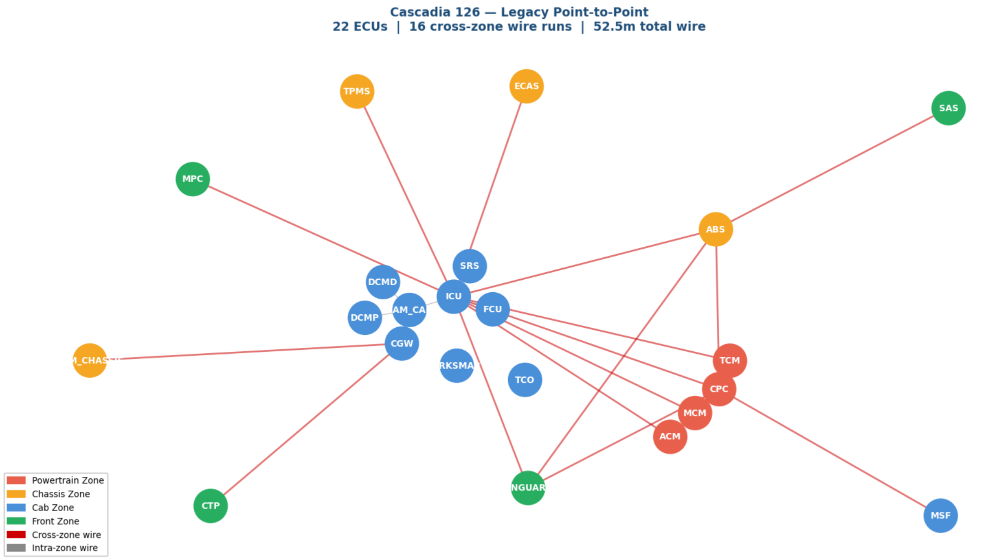
Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	ECM, TCU	2
Chassis Zone	CHZ	ZC-CH	ABS_ESP, EPS	2
Cab Zone	CBZ	ZC-CAB	IC, BCM, HVAC, ACM_AIR, TCO	5
Front Zone	FRZ	ZC-FR	FCM, LDW, TLM	3

Table 3.3: Zone layout — Mercedes-Benz Sprinter 519 CDI

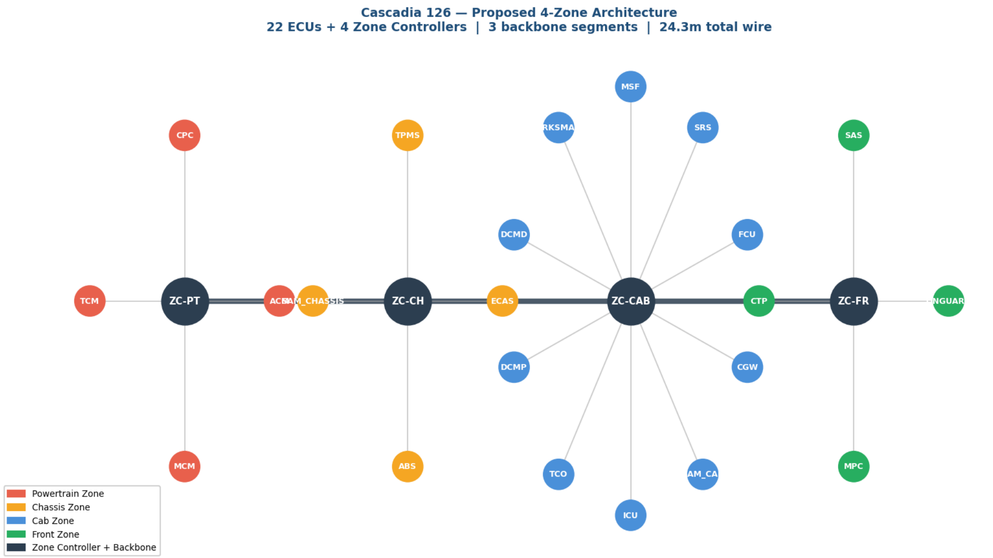
4. Network Topology Diagrams — All Three Vehicles

Each vehicle is shown twice: first in its current legacy wiring layout, then as it would look under the proposed 4-zone Ethernet architecture. Node colour indicates the physical zone. Red lines are cross-zone wire runs. In the zonal diagrams, dark hub nodes are Zone Controllers connected by the Ethernet backbone.

4.1 Freightliner Cascadia 126

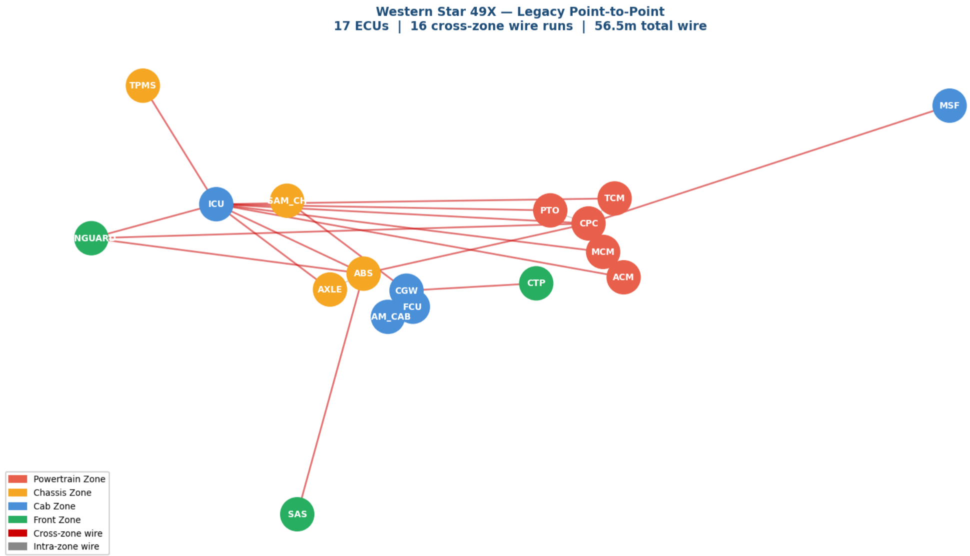


Legacy: 22 ECUs | 16 cross-zone wire runs | 52.5m total wire

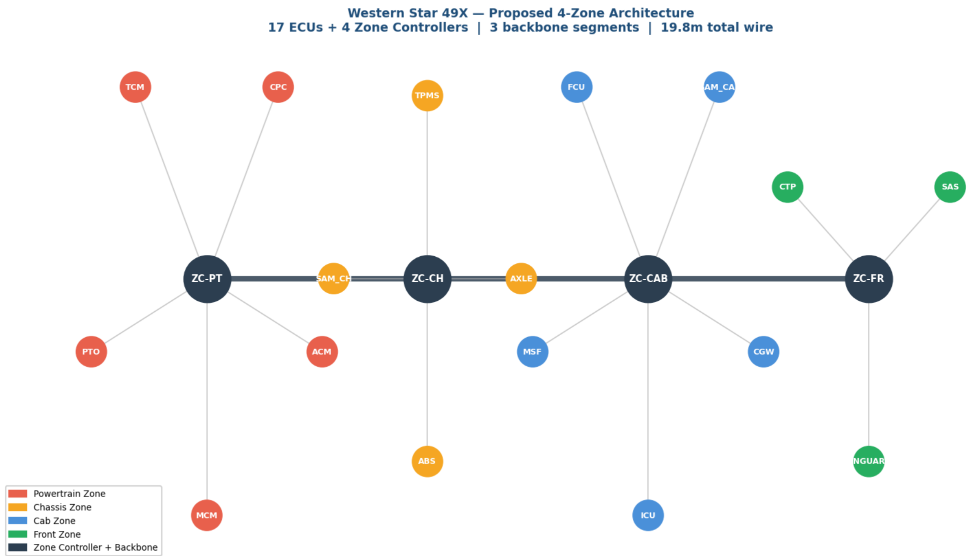


Zonal: 22 ECUs + 4 Zone Controllers | 3 backbone segments | 24.3m total wire | 53.7% reduction

4.2 Western Star 49X

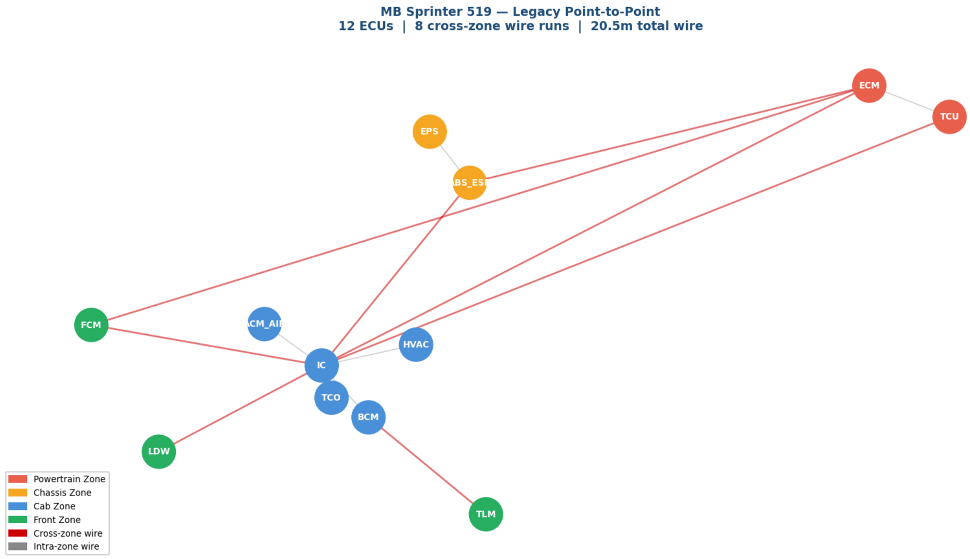


Legacy: 17 ECUs | 16 cross-zone wire runs | 56.5m total wire

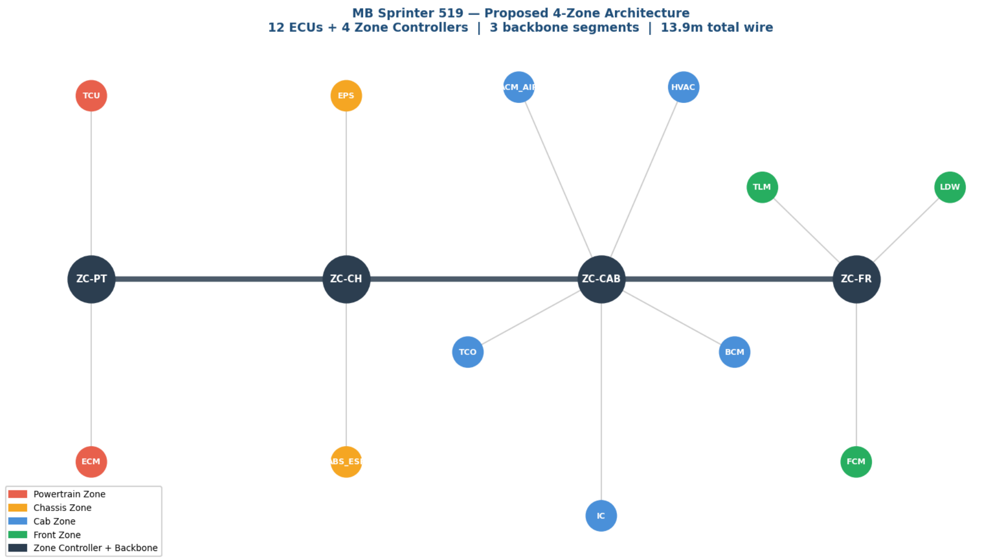


Zonal: 17 ECUs + 4 Zone Controllers | 3 backbone segments | 19.8m total wire | 65.0% reduction

4.3 Mercedes-Benz Sprinter 519 CDI



Legacy: 12 ECUs | 8 cross-zone wire runs | 20.5m total wire



Zonal: 12 ECUs + 4 Zone Controllers | 3 backbone segments | 13.9m total wire | 32.2% reduction

5. Harness Reduction Analysis

The following analysis quantifies the wire length, cross-zone connection, and weight reduction achieved by switching from legacy to zonal architecture. The same ECUs and same logical communication connections are used in both cases — only the wiring topology changes.

5.1 Freightliner Cascadia 126

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	52.5 m	24.3 m	**53.7%**
Cross-zone wire runs	16	3	**81.2%**
Est. harness weight	6.3 kg	2.9 kg	**3.4 kg saved**

Table 5.1: Harness metrics — Freightliner Cascadia 126

5.2 Western Star 49X

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	56.5 m	19.8 m	**65.0%**
Cross-zone wire runs	16	3	**81.2%**
Est. harness weight	6.8 kg	2.4 kg	**4.4 kg saved**

Table 5.2: Harness metrics — Western Star 49X

5.3 Mercedes-Benz Sprinter 519 CDI

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	20.5 m	13.9 m	**32.2%**
Cross-zone wire runs	8	3	**62.5%**
Est. harness weight	2.5 kg	1.7 kg	**0.8 kg saved**

Table 5.3: Harness metrics — Mercedes-Benz Sprinter 519 CDI

The reduction in wire length from 52.5m to 24.3m is driven by the elimination of cross-zone runs, particularly between the Powertrain Zone (PTZ) and the Chassis Zone (CHZ). The zonal layout has reduced the number of connections between zones, resulting in a more localized wiring architecture. For example, the connections between the MCM, ACM, and CPC in the PTZ are now more contained, reducing the need for lengthy wiring.

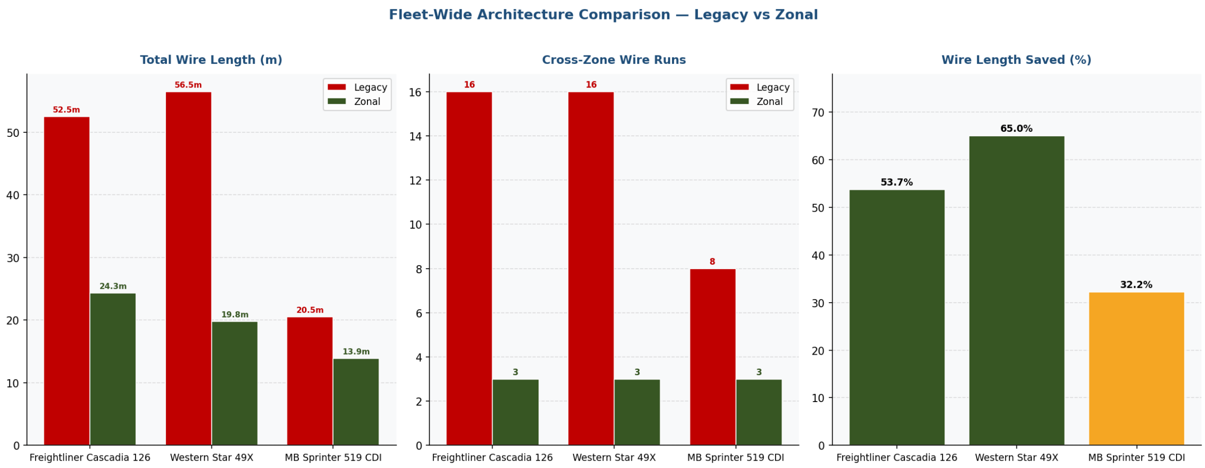


Figure A: Fleet-wide comparison. Wire length (left), cross-zone runs (centre), percentage savings (right). Red = legacy, green = zonal.

6. Communication Hub Analysis

Some ECUs communicate with many others and become natural hubs. Understanding which ECUs are hubs matters because: (1) they determine how much traffic each Zone Controller must handle, and (2) a hub ECU failing could disconnect many other ECUs — making it a single point of failure.

The top 3 communication hubs are the ICU with 12 connections, the CPC with 6 connections, and the ABS with 4 connections. A high connection count indicates a higher load on the zone controller, which can impact performance and reliability. The ICU, with its 12 connections, is a single point of failure, as it is a critical node in the Cab Zone (CBZ). If the ICU fails, it could disrupt communication between multiple ECUs, including the SAM_CAB, CGW, and FCU.

6.1 Full Connection Map — Freightliner Cascadia 126

From ECU	To ECU	What data flows
MCM	ICU	Engine RPM, coolant temp, oil pressure -> dashboard gauges
CPC	ICU	Vehicle speed, fuel level, DPF status -> dashboard
ACM	ICU	DEF level, aftertreatment warnings -> dashboard
TCM	ICU	Gear position, transmission temp -> dashboard
MCM	CPC	Engine torque, speed data
CPC	TCM	Engine torque request for gear shift coordination
MCM	ACM	Exhaust temp, engine load -> aftertreatment control
ABS	ICU	ABS active warning, ESC intervention status
ABS	CPC	Wheel speed -> traction control torque reduction
ABS	ONGUARD	Wheel speed, brake status -> collision avoidance
ECAS	ICU	Suspension height, leveling status
TPMS	ICU	Tire pressure warnings per axle
SAS	ABS	Steering angle -> electronic stability control input
DCMD	SAM_CAB	Driver door ajar, window position
DCMP	SAM_CAB	Passenger door ajar, window position
SAM_CAB	ICU	Door warning status, cab switch states
FCU	ICU	Cab temperature, HVAC mode display
MSF	CPC	Cruise control on/off, speed set commands
SRS	ICU	Airbag fault warning
MPC	ICU	Lane departure alert, camera feed
ONGUARD	ICU	Headway distance, collision warning
ONGUARD	CPC	Emergency brake request
CTP	CGW	OTA data, fleet telemetry uplink
CGW	SAM_CAB	Cross-network signal bridging
CGW	SAM_CHASSIS	Chassis network bridging

Table 6: All 25 connections. In legacy = one dedicated wire each. In zonal = one message route over Ethernet backbone.

7. Zone Boundary Optimality Analysis

Are the zone boundaries drawn in the right place?

Zones are currently drawn based on physical location. But communication patterns do not always follow geography — the engine controller (MCM) talks heavily to the dashboard (ICU) even though they are in different zones.

A graph algorithm called **Greedy Modularity Community Detection** was applied to find the mathematically optimal zone groupings based purely on who talks to whom — ignoring physical location entirely. The result is compared to the human-designed zones using a score called **Modularity** (range 0–1, higher is better).

Design Approach	Zone Count	Modularity Score	What this means
Human-defined (physical location)	4	**0.028**	Current real-world zone layout
Algorithm-optimal (communication pattern)	5	**0.4024**	Best possible grouping by algorithm

Table 7: Zone boundary analysis. Zone efficiency score = 7.0% (human modularity / optimal modularity x 100).

The 7.0% zone efficiency score indicates that the current zonal layout is not optimal, with opportunities for improvement. The physical and communication-optimal zones differ due to the complexity of the system and the need to balance factors such as wiring length, ECU placement, and communication protocols. The MCM and ABS are flagged as boundary nodes because they have connections to multiple zones, making them critical to the overall system architecture. These ECUs are likely to be part of multiple communication paths, increasing their importance in the system.

7.1 ECUs Flagged as Communication Boundary Nodes

These ECUs are physically in one zone but communicate primarily with ECUs in other zones. They are not in the wrong physical location — they are flagged because their communication pattern crosses zone boundaries.

ECU	Physical Zone	Algorithm Cluster	Main communication partners
MCM	Powertrain Zone	Cluster 1	ICU, CPC, ACM
ACM	Powertrain Zone	Cluster 1	ICU, MCM
CPC	Powertrain Zone	Cluster 2	ICU, MCM, TCM, ABS, MSF, ONGUARD
TCM	Powertrain Zone	Cluster 2	ICU, CPC
ABS	Chassis Zone	Cluster 2	ICU, CPC, ONGUARD, SAS
ECAS	Chassis Zone	Cluster 1	ICU
TPMS	Chassis Zone	Cluster 1	ICU
SAM_CHASSIS	Chassis Zone	Cluster 3	CGW
MSF	Cab Zone	Cluster 2	CPC
MPC	Front Zone	Cluster 1	ICU
CTP	Front Zone	Cluster 3	CGW

Table 8: Boundary ECUs. Not misplaced physically — flagged because communication partners span multiple zones.

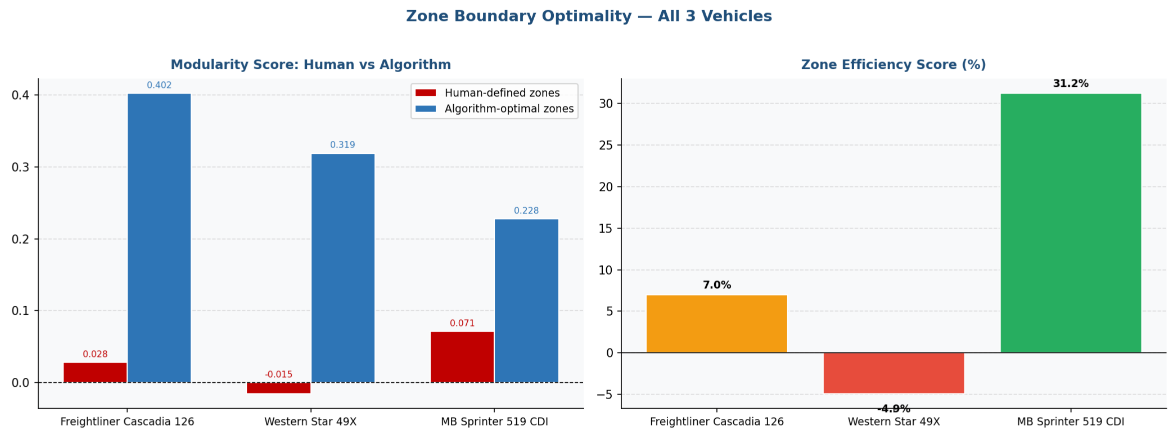


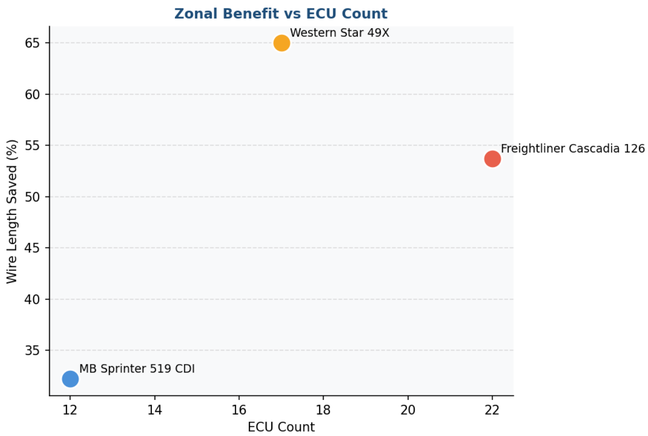
Figure B: Zone optimality across all 3 vehicles. Left: modularity scores. Right: zone efficiency scores. All vehicles show low efficiency — confirming this physical-vs-communication tradeoff is structural, not vehicle-specific.

8. Fleet-Wide Comparative Analysis

Running the same analysis across three different vehicles reveals how zonal architecture benefit scales with vehicle size and complexity.

Vehicle	ECUs	Connections	Legacy Wire (m)	Zonal Wire (m)	Wire Saved	Weight Saved	Zone Efficiency
Freightliner Cascadia 126	22	25	52.5m	24.3m	**53.7%**	3.4kg	7.0%
Western Star 49X	17	22	56.5m	19.8m	**65.0%**	4.4kg	-4.9%
Mercedes-Benz Sprinter 519 CDI	12	13	20.5m	13.9m	**32.2%**	0.8kg	31.2%

Table 9: Fleet-wide comparison — all three vehicles.



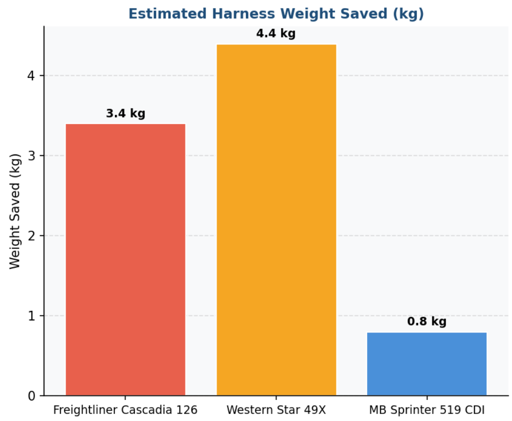


Figure C (left): Wire saved vs ECU count — savings do not scale linearly; physical spread matters more than ECU count. Figure C (right): Absolute harness weight saved per vehicle.

The Western Star 49X benefits most from the zonal architecture, with a 65.0% reduction in wire length and a higher ECU count threshold of 17. In contrast, the Mercedes-Benz Sprinter 519 CDI has a lower ECU count of 12 and a lower wire reduction of 32.2%. The zone efficiency scores vary across the fleet, with the Mercedes-Benz Sprinter 519 CDI having a score of 31.2%, indicating a more optimal zonal layout. The Freightliner Cascadia 126 has a lower zone efficiency score of 7.0%, suggesting opportunities for improvement.

8.1 Key Cross-Fleet Findings

Finding 1: Zonal wire savings do not scale linearly with ECU count. Physical spread of ECUs across the chassis matters more than total ECU count.

Finding 2: Cross-zone wire runs always collapse to exactly 3 backbone segments in a 4-zone design regardless of vehicle size — a structural property of the topology.

Finding 3: Zone efficiency scores are consistently low (7.0%, -4.9%, 31.2%) across all vehicles — confirming the physical-vs-communication tradeoff is fundamental to automotive E/E architecture.

Finding 4: The Sprinter van (12 ECUs) still saves 32.2% wire length. However, adding 4 zone controllers on a 12-ECU platform adds proportionally more hardware overhead — suggesting a practical ROI threshold of ~15+ ECUs.

9. AI-Assisted Engineering Review

The following engineering review was generated by Groq Llama 3.3 70B based solely on the computed data above. The model was instructed to act as a senior automotive E/E architect and provide specific, physically valid recommendations.

9.1 Design Recommendations

1. The MCM should be moved from the Powertrain Zone (PTZ) to the Chassis Zone (CHZ) to reduce cross-zone runs and improve the overall system architecture.

2. The zone controller for the Cab Zone (CBZ) should be placed within 1.5m of the ICU to reduce wiring length and improve communication efficiency.

3. A redundant communication path should be implemented for the ICU to ensure that critical communication is maintained in the event of a failure, such as a backup connection to the SAM_CAB or CGW.

9.2 Analysis Limitations

• The analysis is limited by the assumption that the current zonal layout is the only possible configuration, and alternative layouts may yield better results.

• The model does not account for potential electromagnetic interference (EMI) or other environmental factors that may impact the system's performance and reliability.

10. Methodology and Data Sources

10.1 How the Tool Works

Step 1 — Input (YAML config file): Each vehicle's ECUs, zone assignments, and connections are described in a YAML text file. Wire length estimates are based on published chassis dimensions.

Step 2 — Graph building (graph.py): Two NetworkX graphs are built: legacy (ECU-to-ECU edges weighted by wire length) and zonal (ECU-to-ZC stubs + backbone).

Step 3 — Metrics (metrics.py): Total wire length, cross-zone run count, and harness weight are computed by summing edge weights.

Step 4 — Zone optimality (optimizer.py): Greedy Modularity Community Detection finds communication-optimal zone groupings. Modularity score and zone efficiency are calculated.

Step 5 — Fleet comparison (fleet_compare.py): Steps 1-4 run for all three vehicles and results are compiled into cross-vehicle tables.

Step 6 — AI review (reviewer.py): Computed data is passed to Groq Llama 3.3 70B with a structured automotive engineering prompt.

Step 7 — PDF generation (build_pdf_report.py): Matplotlib generates all diagrams. ReportLab assembles them with tables and text into this PDF.

10.2 Data Sources

1. DTNA SS-1033423 — J-1939 Fault Code Source Address Descriptions. Daimler Trucks North America. NHTSA public database. Primary ECU list source for Cascadia 126.

2. DTNA STI-503 — NGC sSAM, VPDM & BCA Wall Chart. Rev. Q, March 2020. Physical ECU location confirmation.

3. Western Star Bodybuilder Manual Rev 3.1 — Western Star 49X ECU configuration.

4. Mercedes-Benz XENTRY public diagnostic documentation — Sprinter 519 CDI ECU configuration.

5. Park C, Cui C, Park S. Analysis of E2E Delay and Wiring Harness in Zonal Architecture. Sensors. 2024;24(10):3248. Academic methodology validation.

6. SAE J1939 — Serial Control and Communications Heavy Duty Vehicle Network.

10.3 Key Assumptions

- Wire lengths estimated from published chassis dimensions: Cascadia 126 ~8.5m, Western Star 49X ~7.5m, Sprinter ~5.5m
- Harness weight calculated using 120 g/m bundled average — conservative estimate for signal-wire harnesses
- Analysis covers signal and data wires only — power distribution harness excluded
- Static ECU configuration — optional equipment and variant builds not modelled
- All ECU data sourced from public DTNA and Mercedes-Benz documents — no proprietary OEM data used

Zonal E/E Architecture Analyzer | Open-source personal R&D; project | https://github.com/Rish-736/zonal-ee-analyzer
Built alongside DTICI internship research — Rishit, ECE '28, VIT Vellore — June 2026